PART 1 EXECUTIVE SUMMARY AND CORE ARCHITECTURAL BRIEF

Self State Aware Autonomous Execution

A resource-accountable execution architecture for Google DeepMind

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At the invitation of Dr. John I. Glass of the J. Craig Venter Institute (JCVI), our joint research on human–AI collaboration is scheduled for presentation at the 6th Minimal Cell Virtual Workshop on 21 October 2026, with Jeongin Choi presenting.

We seek a three-month Google DeepMind evaluation of an execution layer that authorizes agent actions against current evidence and resource limits. The proposed sandboxed PoC offers a fixed budget and reproducible deliverables.

The problem and proposed contribution

A plausible action may rely on stale evidence, unavailable capabilities, or an exhausted budget. The proposed architecture couples state, historical residue, resource accounting, and authorization in one inspectable state, with an independently enforced gate between proposal and execution.

$$S(t) = (X(t),\ R(t),\ B(t),\ Q(S))$$

Element	Architectural responsibility
X(t)	Current system state, owned capabilities, observations, and unresolved conditions.
R(t)	History and damage ledger; retain events and the evidence behind state changes.
B(t)	Resource ledger covering energy or compute, time, tool calls, and reserves.
Q(S)	Binary execution gate; authorize only after all required checks succeed.

Execution policy

For each proposed action, the monitor checks evidence freshness, capability and safety preconditions, and a conservative cost bound including its own overhead. A failed or unresolved check sets the gate to zero for that action. The agent records OPEN and invokes an agreed fallback. A separate watchdog bounds inference and tool use. Q = 0 never grants the model authority to disable the monitor or independently select a destructive shutdown.

Verification and auditability statement

The architecture defines an explicit, inspectable execution policy coupling current state X(t), residue R(t), resource ledger B(t), and binary gate Q(S). Automated checks in the supplied deterministic symbolic simulation confirm reserve-preserving debits for accepted actions and unchanged resources for denied actions. Scenario assertions confirm rejection of stale proposals and selection of modeled recovery or refuge paths. These results apply to the declared capability and cost model. LLM action grounding, independent watchdogs, real monitoring costs, and prevention of irreversible effects remain requirements for the proposed PoC.

Proposed research support is USD 15,000 across three milestones, plus up to USD 1,500 for agreed API costs. Existing background IP stays with its current owners. Successful completion of the agreed PoC acceptance criteria would trigger the proposed follow-on terms: a principal advisory agreement and Google-funded private research office. The requested next step is a 20-minute scoping discussion with a nominated technical reviewer.